Digital Addiction Analysis

Using Python, SQL, and Power BI to study screen time and sleep

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Introduction

Problem

Rising smartphone & social media use → impacts sleep, mental health

Goal

Analyze behavior and patterns linking screen time to sleep

Approach

Data analytics pipeline: Python → SQL → Power BI

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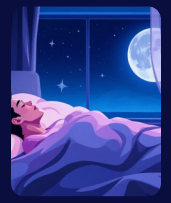
Problem Statement

Excessive Screen Time

High daily usage across users



Reduced Sleep

Lower sleep duration linked to device use

Young Users

Higher dependency among younger age groups

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Objectives

Screen Time

Analyze daily screen time behavior

Sleep Impact

Study screen time vs sleep duration

Age Groups

Identify most affected demographics

Dashboard

Build interactive Power BI visualizations

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Dataset Description

- **Fields**

  user_id, age, gender, screen_time_hours, sleep_hours, other behaviors

- **Format**

  CSV used across Python, SQL, Power BI

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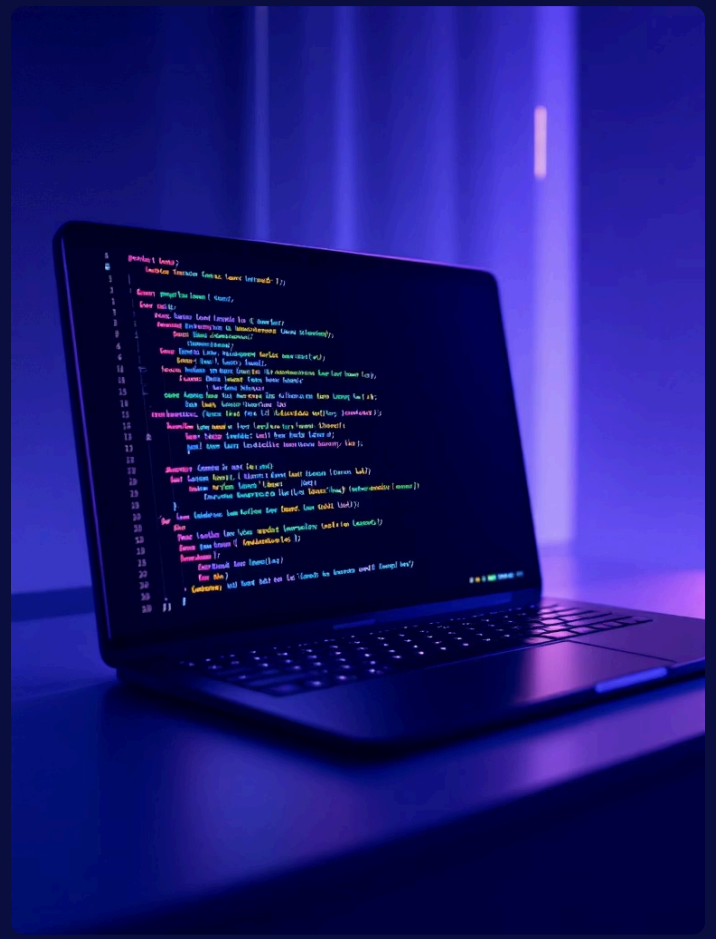

Data Preprocessing (Python)

- Loaded with Pandas
- Checked structure & types
- Identified missing values

Ensured clean dataset for analysis

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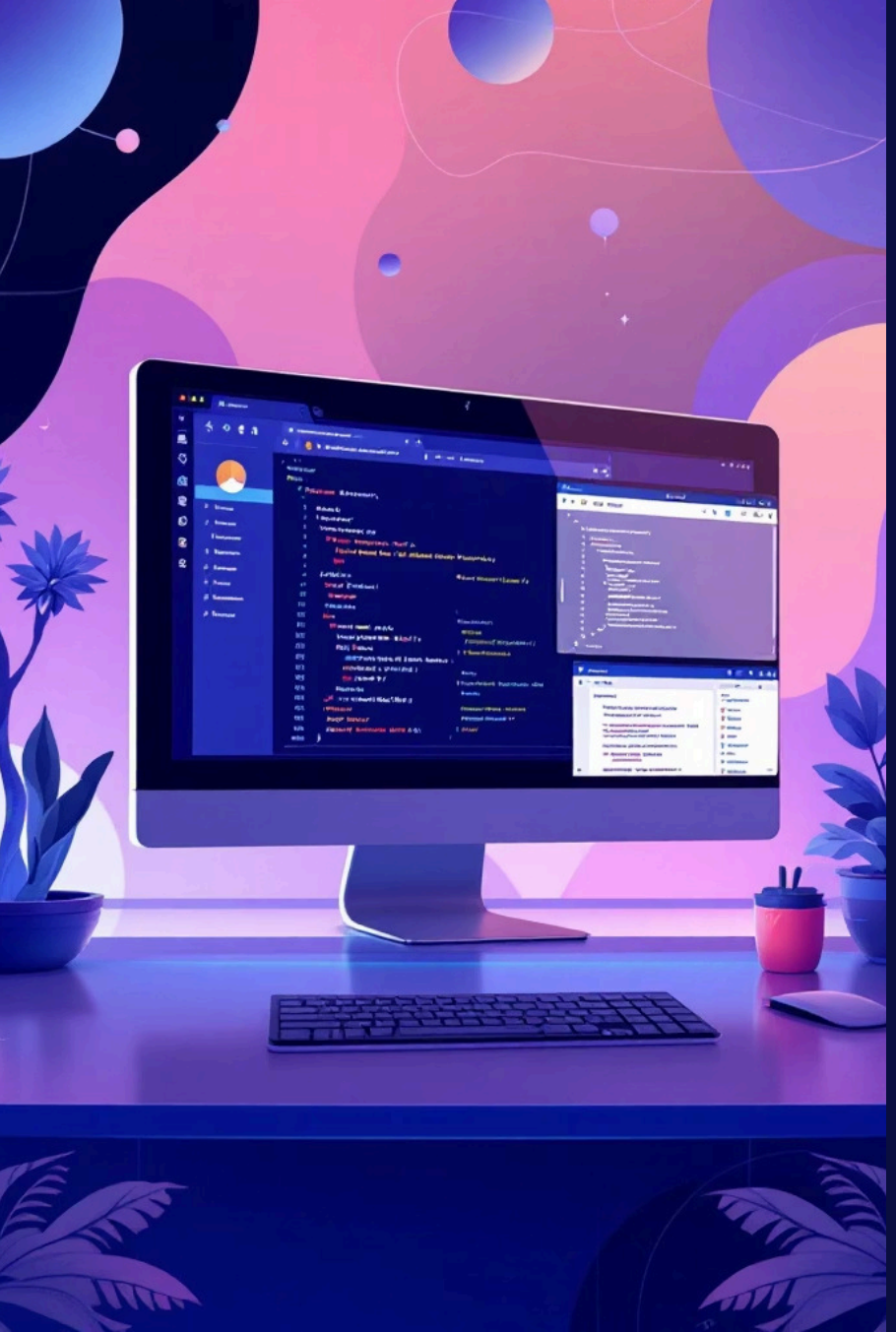

SQL Analysis

Total Users

Count users in dataset

Avg Screen Time

Compute average daily usage

High Usage

Number of high screen-time users

Gaming by Gender

Avg gaming hours per gender

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Power BI Dashboard

KPI Cards

Total users, avg screen time, high users

Scatter Plot

Screen time vs sleep hours

Donut Chart

Users by age group

Clustered Columns

Avg sleep hours by screen-time

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Key Insights

Sleep Impact

Higher screen time → lower sleep duration

Demographics

Younger age groups show higher engagement

Pattern

Excessive usage indicates potential addiction

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Conclusion & Future Scope

Conclusion

Analytics show link between screen time and sleep; dashboard conveys insights

Future

Add mental health indicators; predictive models; larger or real-time data

Outcome

Data-driven understanding useful for research and decisions

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